Total No. of Questions : 4+4] [Total No. of Printed Pages : 7

Roll No. ____________________

BS/Sem IV/107 (AB)

B. Sc. (Hons) Semester IV Examination
2 0 1 3 – 1 4

CHEMISTRY

Paper No. CHB-401 : Inorganic Chemistry–II and Organic Chemistry–II

Time : 3 hours **Full Marks : 70**

(Write your Roll No. at the top immediately on the receipt of this question paper)

The figures in the right-hand margin indicate marks.
USE SEPARATE ANSWER BOOK FOR EACH SECTION

Section—A
(Inorganic Chemistry–II)
(Marks : 35)

*Answer **three** questions including Question 1 which is compulsory*

1. 1. What happens when CH_3COOH is dissolved in liquid NH_3 and H_2SO_4? Explain. **3**

2. Why is alkali-metal liquid NH_3 solution deep blue and good reducing agent? **2**

3. Why are Ti^{3+} complexes coloured while Ti^{4+} complexes are colourless? **2**

4. Why is decrease in size from Ce to Lu very small? Explain. **2**

5. Which one is a better Lewis acid BCl_3 or $B(CH_3)_3$? **2**

2. 1. Identify the Brønsted-Lowry acid, base, conjugate acid and conjugate base in the following : **3**

(i) $NH_4^+ + S^{2-} \rightleftharpoons HS^- + NH_3$

(ii) $CO_3^{2-} + H_2O \rightleftharpoons HCO_3^- + OH^-$

2. What are the characteristic species of acids and bases in BrF_3 and liquid NH_3? **2**

3. Giving three equations, illustrate Lewis acids and bases. **3**

4. Complete the reactions and identify the Lux-Flood acids and bases in the following : **2**

(i) $Na_2O + SiO_2 =$

(ii) $3Na_2O + P_2O_5 =$

5. In terms of HSAB principle which end of NCO^- would you expect to coordinate to Co^{2+} and Ti^{4+}? **2**

Q.3. 1. Why do transition metals exhibit variable oxidation states? How many oxidation states are exhibited by Fe and Cr and which one is more common? **3**

2. Write IUPAC names of the following : **3**

   (i) $[Co(ONO)(NH_3)_5]^{2+}$

   (ii) $[(NH_3)_5Cr{-}OH{-}Cr(NH_3)_5]Cl_5$

   (iii) $K_2[Zn(NH_2)_4]$

3. Detect the structure and mention the name of isomers of $[Co(en)_2Cl_2]^+$. **3**

4. Calculate the effective atomic number of the metal in the following : **3**

   (i) $Fe(CO)_5$

   (ii) $[Ni(NH_3)_6]^{2+}$

   (iii) $Co_2(CO)_8$

4. 1. Name some properties that mainly determine the utility of a solvent. **3**

2. Giving two examples compare and contrast the precipitation reactions in aqueous and liquid NH_3. **3**

3. Explain the important consequences of lanthanide contraction. **3**

4. Write outer electronic configuration of Ho and Lu in the ground state and indicate number of unpaired electron in Ho^{3+} and Lu^{3+} ions. **3**

Section—B
(Organic Chemistry–II)
(Marks : 35)

*Answer **three** questions including Question 1 which is compulsory*

1. 1. Pyridine is more basic than pyrrole but less basic than aliphatic tertiary amines. Explain. **2**

2. Write the cyclic structures of D (+) glucose. **2**

3. How will you prepare indigotin from anthranilic acid? **2**

4. Complete the following reactions : **3×1+2**

   (i) *(Anthracene + Maleic Anhydride system)* $\xrightarrow{\Delta}$?

   (ii) *(Pyridine layout substitution array)* $\xrightarrow[\Delta]{NaNH_2}$?

   (iii) *(Furan gas-phase layout conversions)* $\xrightarrow{HCN\ /\ HCl\ /\ H_2O}$?

   (iv) *(Phthalic anhydride + Benzene architecture)* $\xrightarrow{AlCl_3}$? $\xrightarrow[2.\ Zn]{1.\ c.H_2SO_4}$?

2. 1. Give a reaction sequence to convert glucose into fructose. **4**

2. Write the name of the following reactions and explain the formation of products with suitable mechanism : **3×2**

(i) *(Pyrrole platform conversions via Riemer-Tiemann protocol)* $\xrightarrow{CHCl_3/KOH}$ **A** + **B** $\xrightarrow{\bar{O}H\ /\ H_2O}$?

(ii) *(Pyrrole Vilsmeier-Haack system execution array)* $\xrightarrow[2.\ \bar{O}H\ /\ H_2O]{1.\ DMF\ /\ POCl_3/0\ °C}$?

3. Complete the following reaction : **2**

(i) *(Pyridine ring execution matrices)* $\xrightarrow[2.\ H^+/H_2O]{1.\ KOH}$?

3. 1. Describe Hantzsch synthesis for pyridine. **6**

2. Describe the structure of indigotin. **6**

4. Predict the products of following reactions : **4+(2×2)+4**

(i) Benzene + Succinic anhydride $\xrightarrow{AlCl_3}$? $\xrightarrow{Zn\ /\ Hg\ /\ HCl}$? $\xrightarrow[2.\ 220\ °C]{1.\ NaOH : KOH\ /\ 1:1}$?

(ii) Naphthalene $\xrightarrow{AcOH\ /\ HNO_3}$? $\xrightarrow{KMnO_4/H^+}$?

(iii) Naphthalene $\xrightarrow{O_3}$? $\xrightarrow{Zn\ /\ H_2O}$?

(iv) Aldohexose $\xrightarrow{Br_2/H_2O}$? $\xrightarrow{1.\ CaCO_3}$? $\xrightarrow{2.\ H_2O_2/Fe^{3+}}$?

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